

Test Results Report

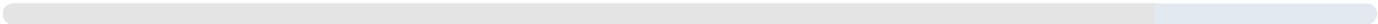
Comprehensive Analysis of RNG Algorithm Testing

Generated: 20 stycznia 2026 10:57

Period: 20 stycznia 2026 09:46 - 20 stycznia 2026 10:56

Executive Summary

TOTAL TESTS 346 10 algorithms tested	PASS RATE 83.8% 290 passed / 56 failed
TEST TYPES 31 Different test types	AVG EXECUTION TIME 12239 ms Per test average



Algorithm Performance Analysis

Comprehensive metrics for all tested RNG algorithms

Algorithm	Pass Rate	Avg Score	Avg Time (ms)	Tests
Python Random	96.7%	0.61	4607	30
System RNG	95.6%	0.47	52576	45
PCG32	95.6%	0.60	5731	45
ChaCha20	90.0%	0.57	5565	30
Park-Miller	90.0%	0.61	4015	30
SplitMix64	90.0%	0.58	3675	30
Xoshiro256**	84.8%	0.49	8750	46
AWCG Standard	83.3%	0.53	3973	30
LCG	76.7%	0.49	4153	30
Blum Blum Shub	23.3%	0.18	14293	30

Test Type Performance Analysis

Metrics for different NIST test types

Test Type	Pass Rate	Avg Score	Avg Time (ms)	Tests
nist_linear_complexity	100.0%	1.00	7643	11
nist_universal	100.0%	1.00	14329	11
nist_non_overlap- ping_template	100.0%	1.00	27865	11
nist_block_frequency	100.0%	0.73	23890	11
frequency_test	100.0%	0.91	45489	1
diehard_opso	100.0%	0.62	5728	12
diehard_binary_rank	100.0%	0.55	4056	12
diehard_runs	91.7%	0.55	4850	12
diehard_overlap- ping_sums	91.7%	0.13	3748	12
diehard_3dspheres	91.7%	0.41	4830	12
diehard_parking_lot	91.7%	0.45	6369	12
diehard_count_1s	91.7%	0.35	6076	12
diehard_dna	91.7%	0.58	2826	12
diehard_oqso	91.7%	0.71	4418	12
nist_cumulative_sums	90.9%	0.91	7695	11
nist_overlap- ping_template	90.9%	0.42	22070	11
nist_runs	90.9%	0.59	33580	11
nist_monobit	90.9%	0.42	24621	11
diehard_craps	83.3%	0.51	7097	12
diehard_bitstream	83.3%	0.63	9837	12
diehard_overlap- ping_permutations	83.3%	0.54	5138	12

Test Type	Pass Rate	Avg Score	Avg Time (ms)	Tests
diehard_birth-day_spacings	83.3%	0.44	18998	12
nist_approximate_entropy	81.8%	0.50	7945	11
nist_serial	81.8%	0.61	7615	11
nist_matrix_rank	81.8%	0.36	30237	11
nist_longest_run	81.8%	0.34	29871	11
nist_random_excursions	72.7%	0.06	7640	11
nist_dft	72.7%	0.22	28425	11
diehard_squeeze	66.7%	0.64	3087	12
nist_random_excursions_variant	36.4%	0.21	7141	11
diehard_minimum_distance	0.0%	0.00	4862	12

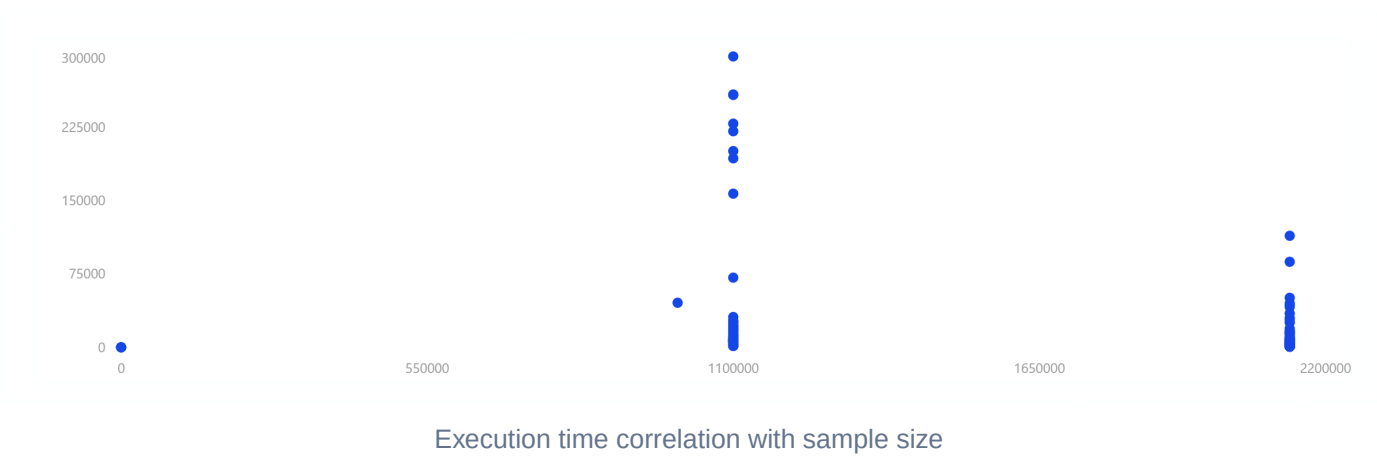
Sample Size Distribution

Analysis of test execution by sample size ranges

Sample Range	Test Count	Avg Time (ms)
0-999	5	0.0
1000000-1000999	1	45488.5
1100000-1100999	160	19288.9
2100000-2100999	180	6127.8



Performance vs Sample Size



Performance Trends by Sample Size

Detailed performance metrics across different sample sizes

Xoshiro256**

Samples	Pass Rate	Avg Score
0k	0.0%	0.00
1000k	100.0%	0.91
1100k	100.0%	0.78
2100k	93.3%	0.47

ChaCha20

Samples	Pass Rate	Avg Score
1100k	93.3%	0.57
2100k	86.7%	0.58

Blum Blum Shub

Samples	Pass Rate	Avg Score
1100k	33.3%	0.33
2100k	13.3%	0.03

AWCG Standard

Samples	Pass Rate	Avg Score
1100k	80.0%	0.46
2100k	86.7%	0.61

System RNG

Samples	Pass Rate	Avg Score
1100k	100.0%	0.57
2100k	93.3%	0.42

Python Random

Samples	Pass Rate	Avg Score
1100k	100.0%	0.66
2100k	93.3%	0.55

PCG32

Samples	Pass Rate	Avg Score
1100k	100.0%	0.63
2100k	86.7%	0.54

Park-Miller

Samples	Pass Rate	Avg Score
1100k	93.3%	0.68
2100k	86.7%	0.54

LCG

Samples	Pass Rate	Avg Score
1100k	80.0%	0.55
2100k	73.3%	0.44

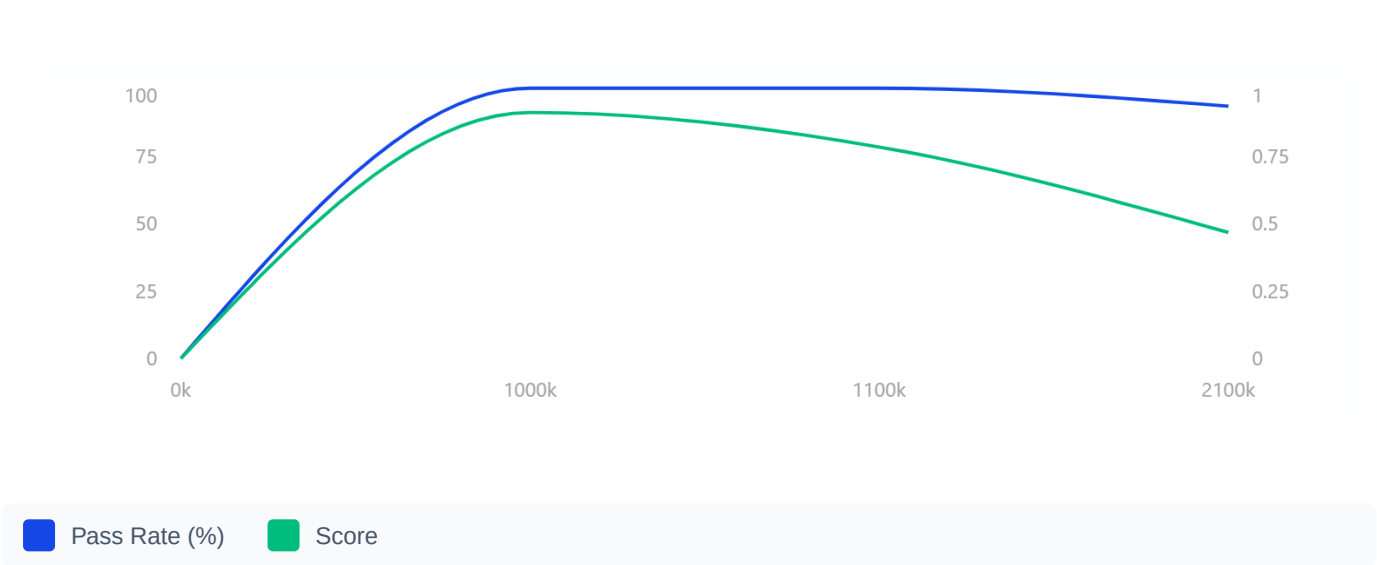
SplitMix64

Samples	Pass Rate	Avg Score
1100k	86.7%	0.53
2100k	93.3%	0.63

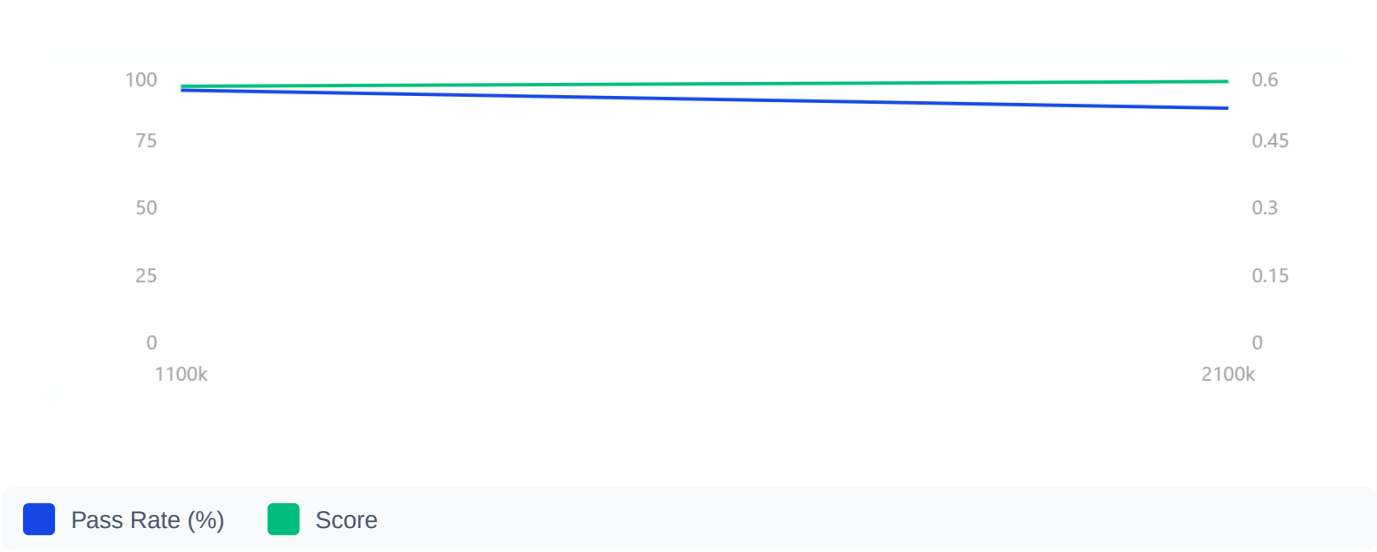
Algorithm Performance Trends

Performance metrics across different sample sizes for each algorithm

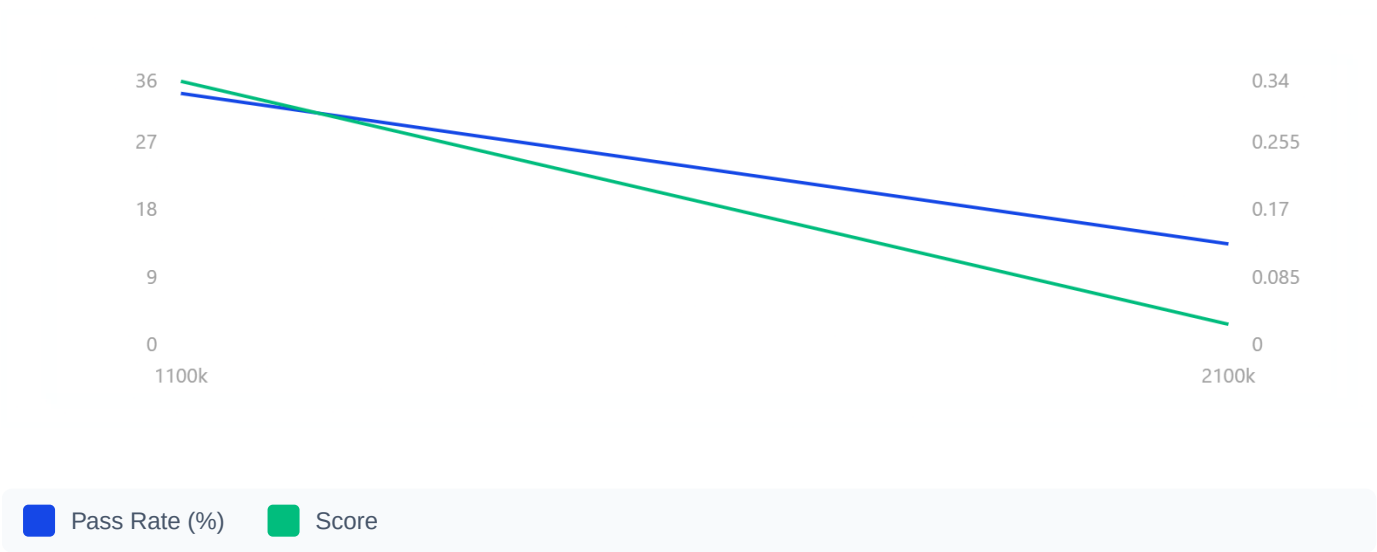
Xoshiro256**



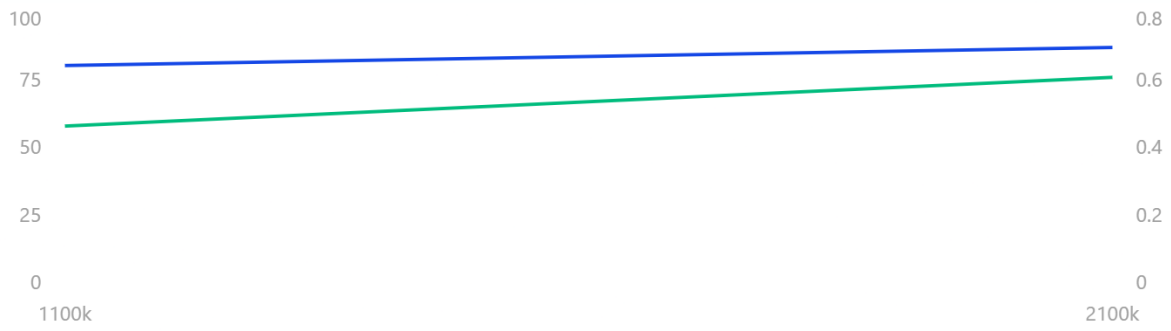
ChaCha20



Blum Blum Shub

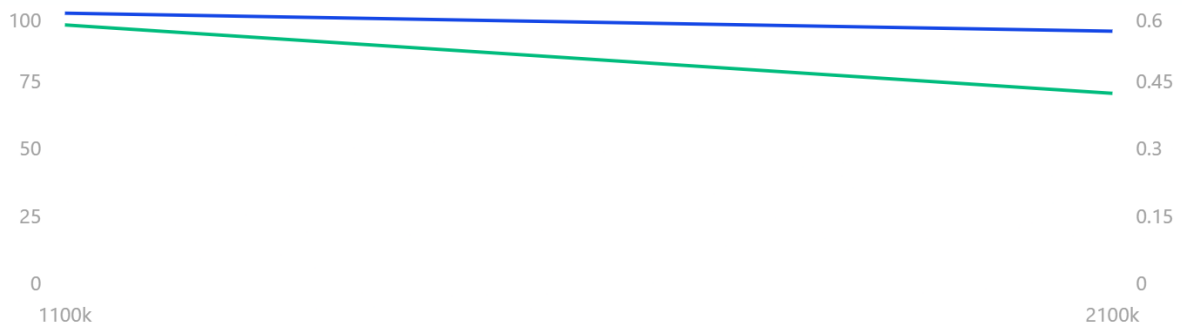


AWCG Standard



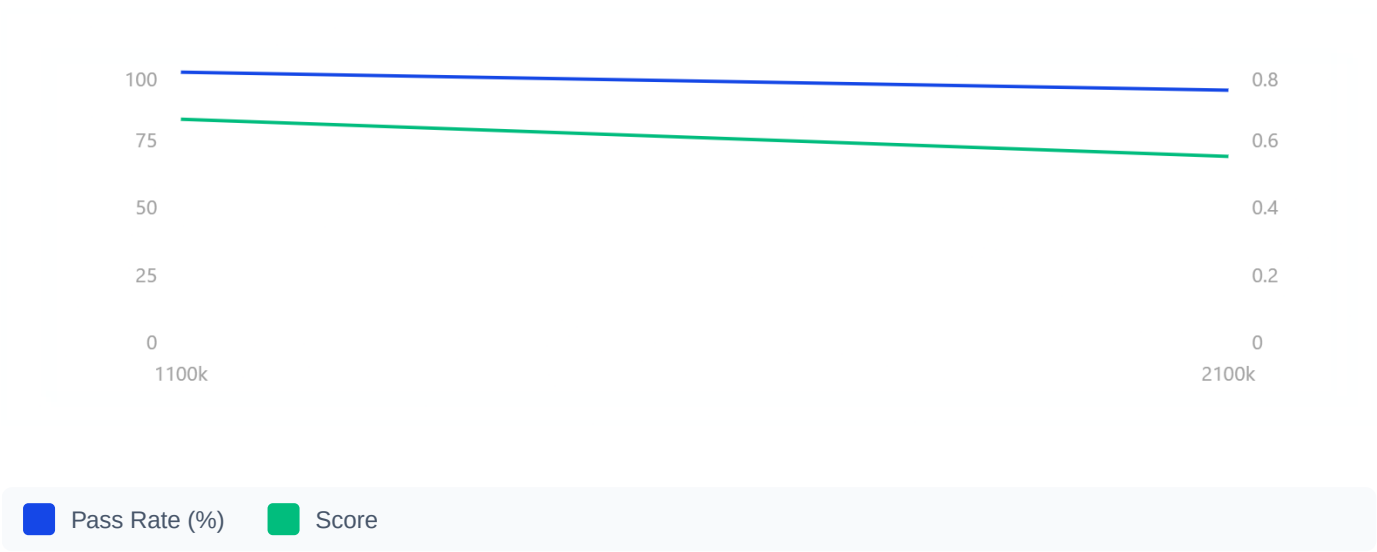
Pass Rate (%) Score

System RNG

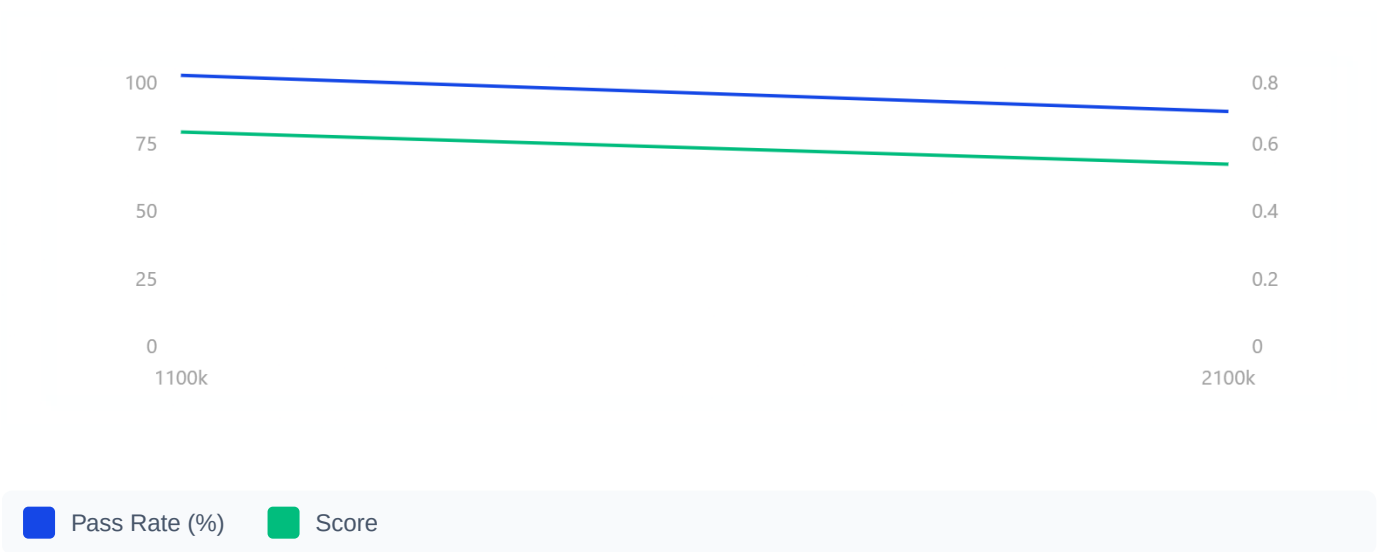


Pass Rate (%) Score

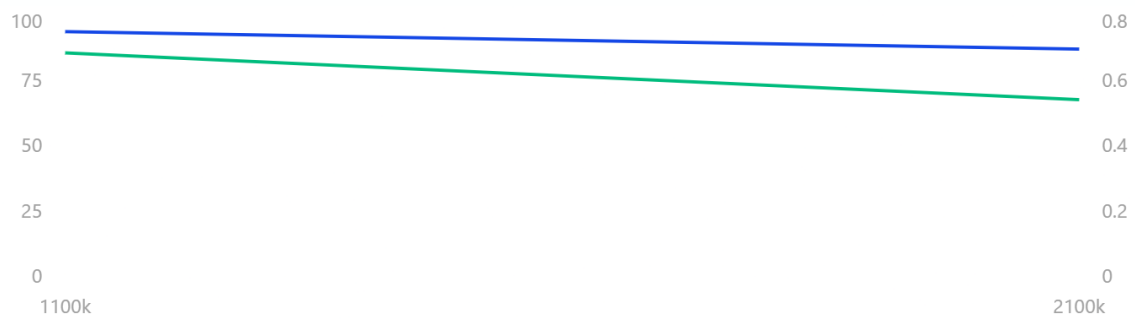
Python Random



PCG32

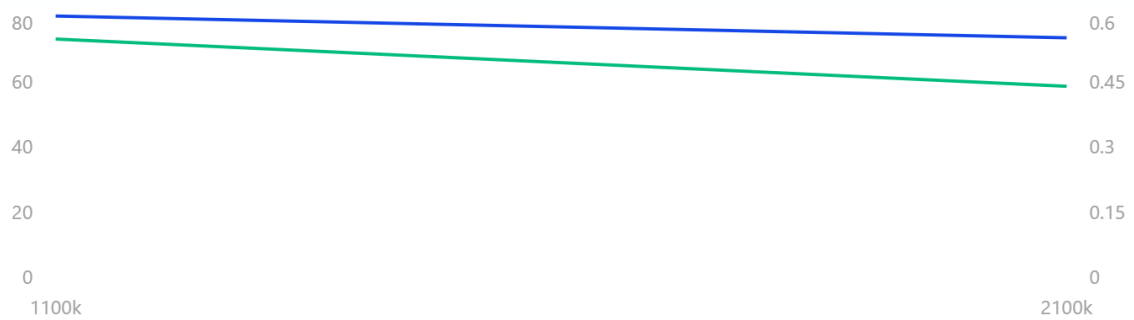


Park-Miller



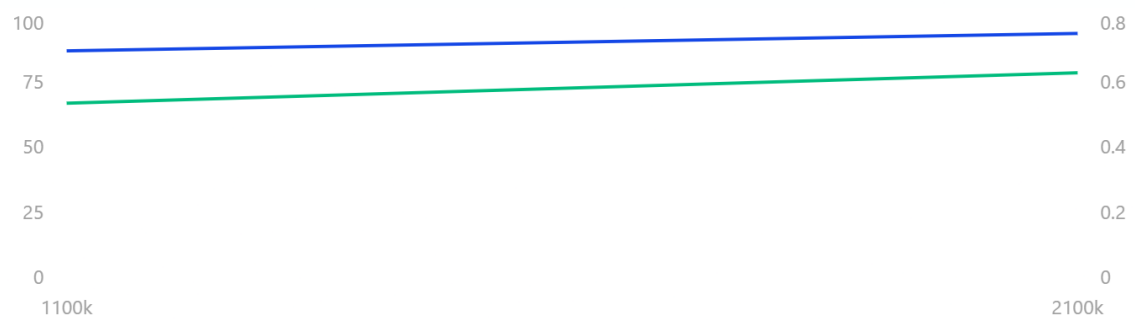
Pass Rate (%) Score

LCG



Pass Rate (%) Score

SplitMix64

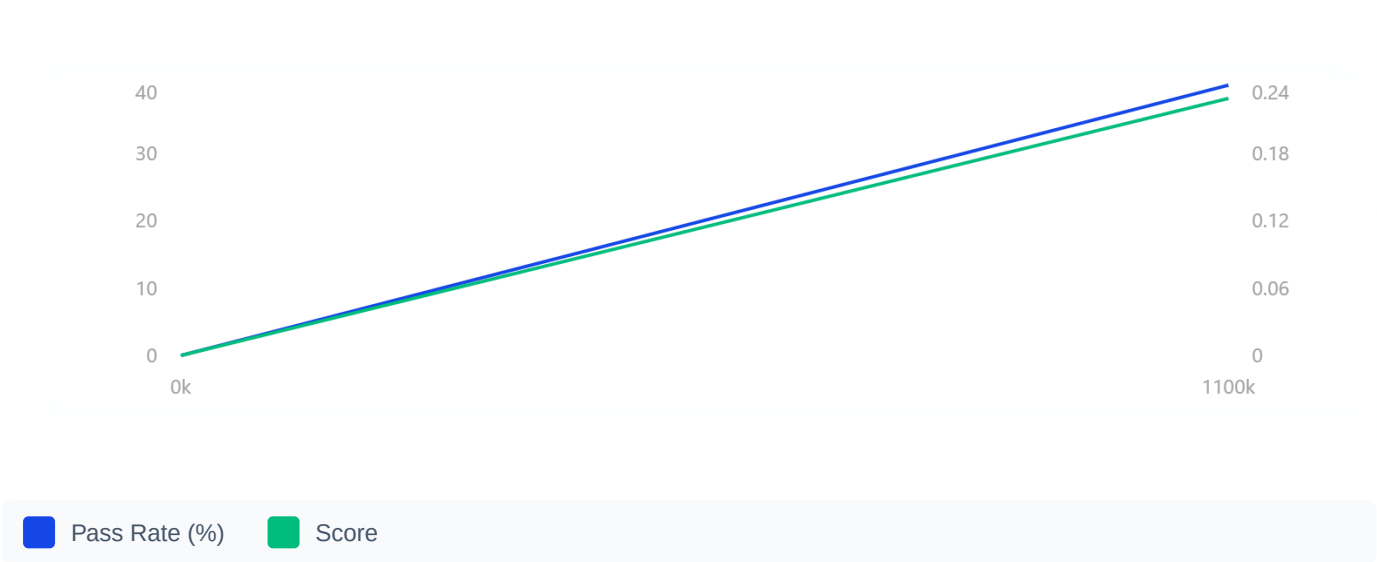


Pass Rate (%) Score

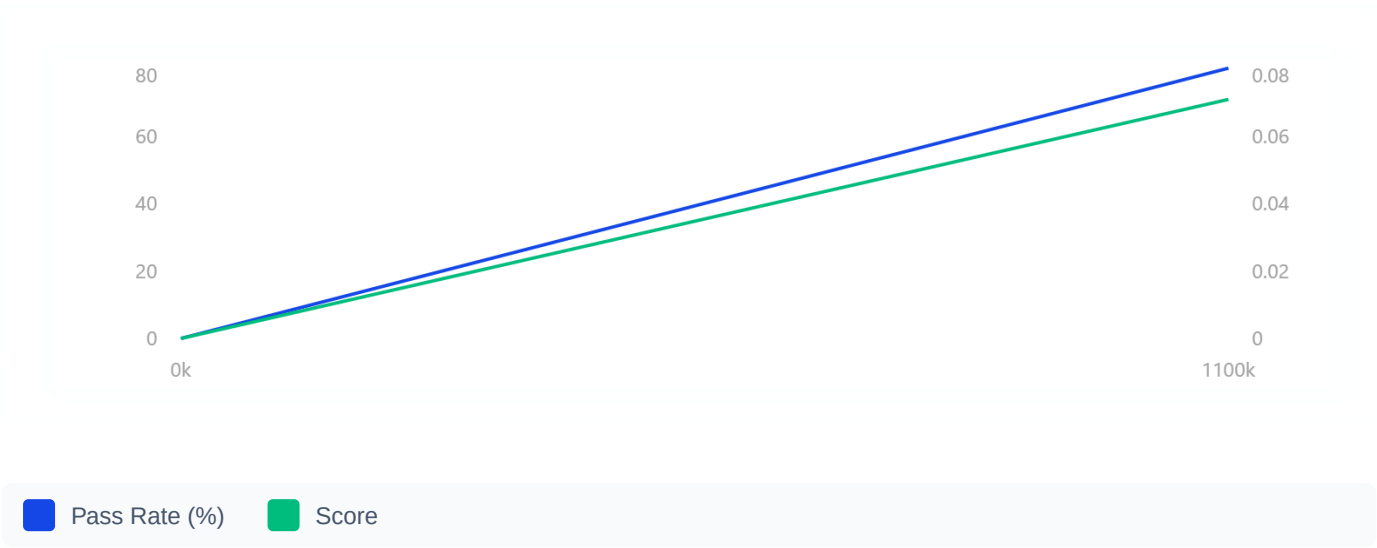
Test Type Performance Trends

Performance metrics across different sample sizes for each test type

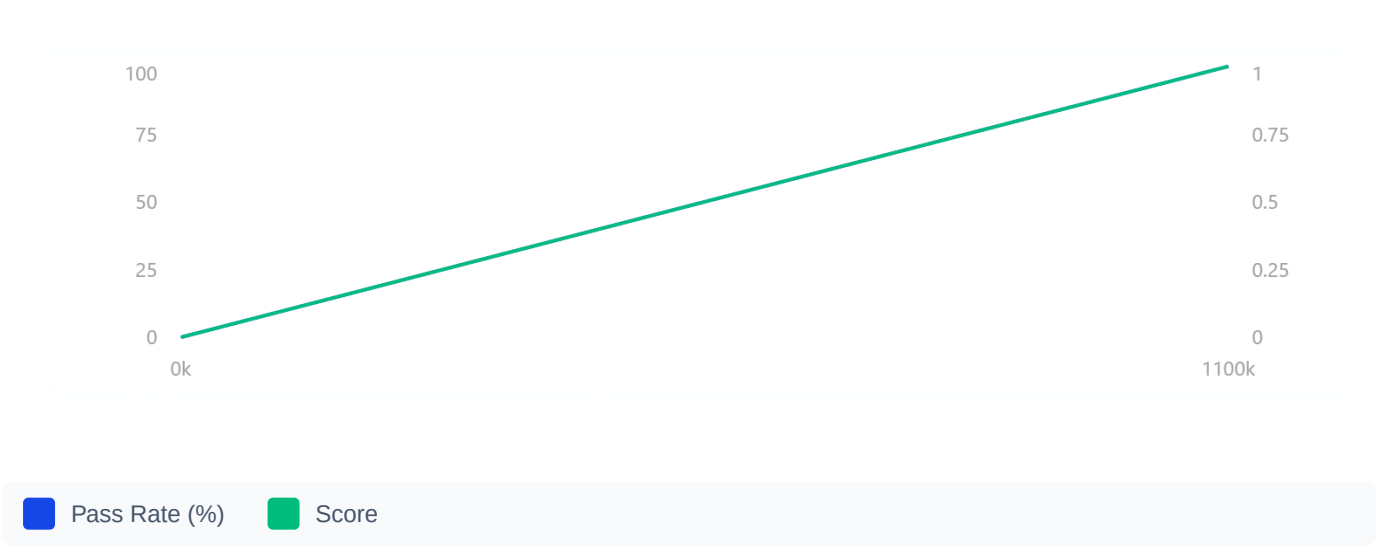
nist_random_excursions_variant



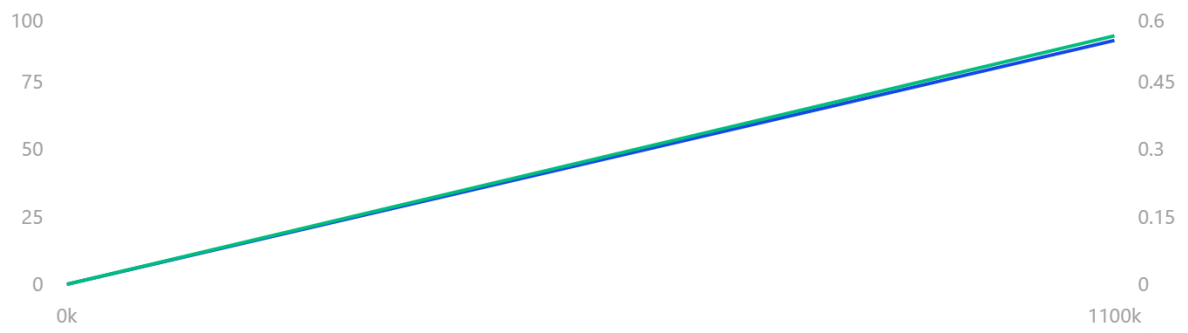
nist_random_excursions



nist_cumulative_sums

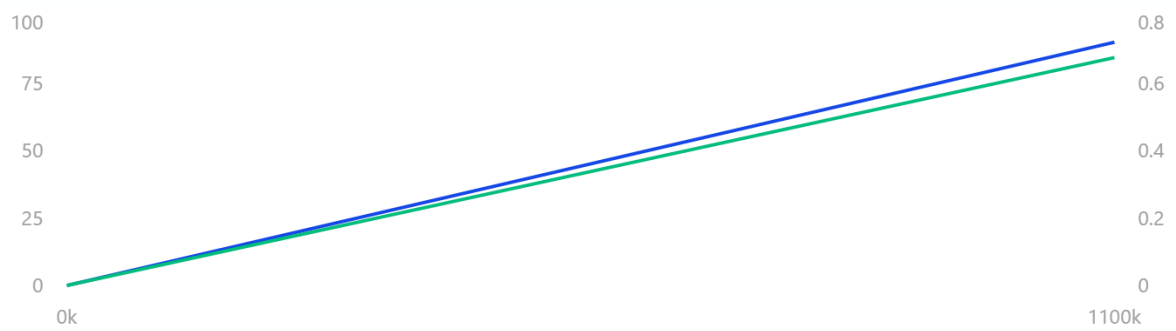


nist_approximate_entropy



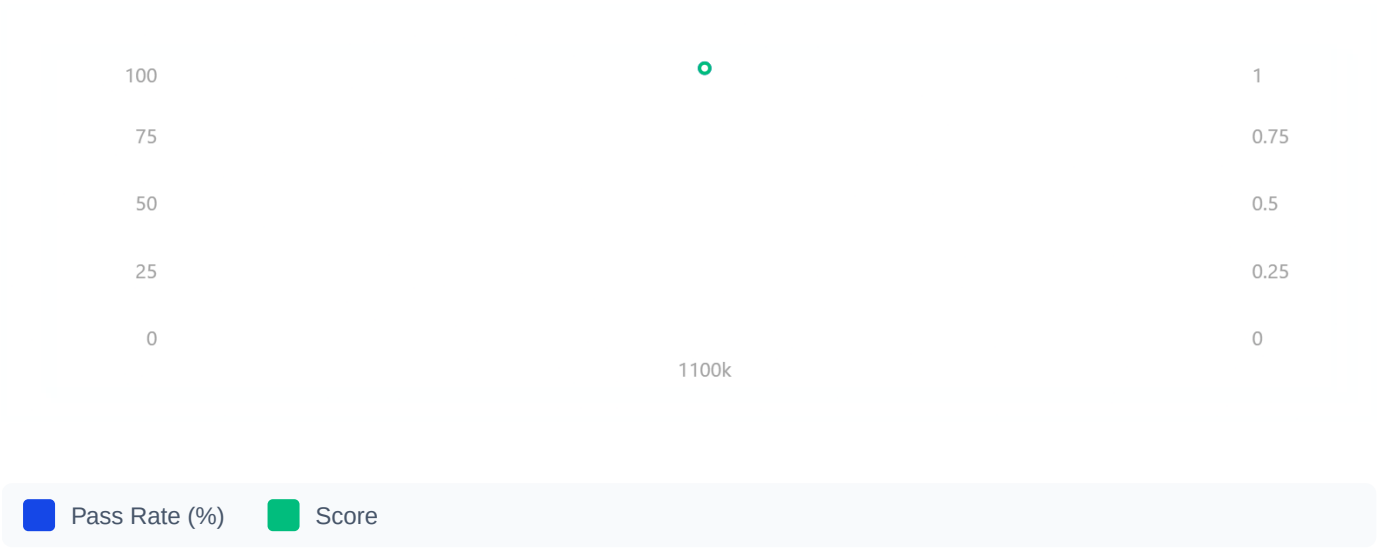
Pass Rate (%) Score

nist_serial

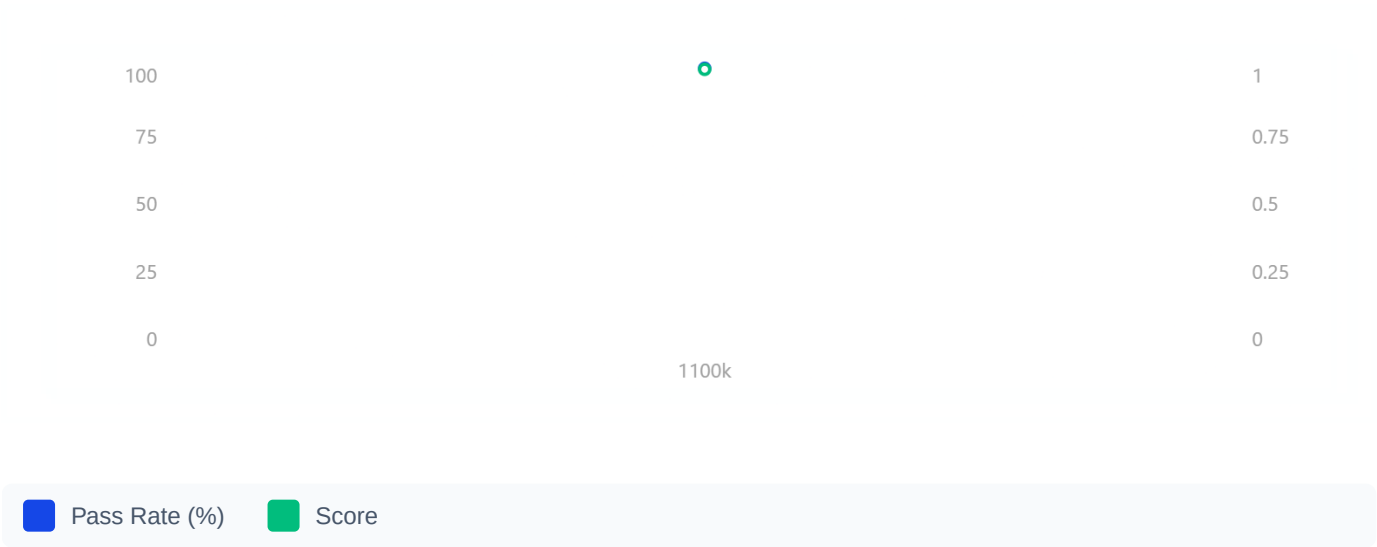


Pass Rate (%) Score

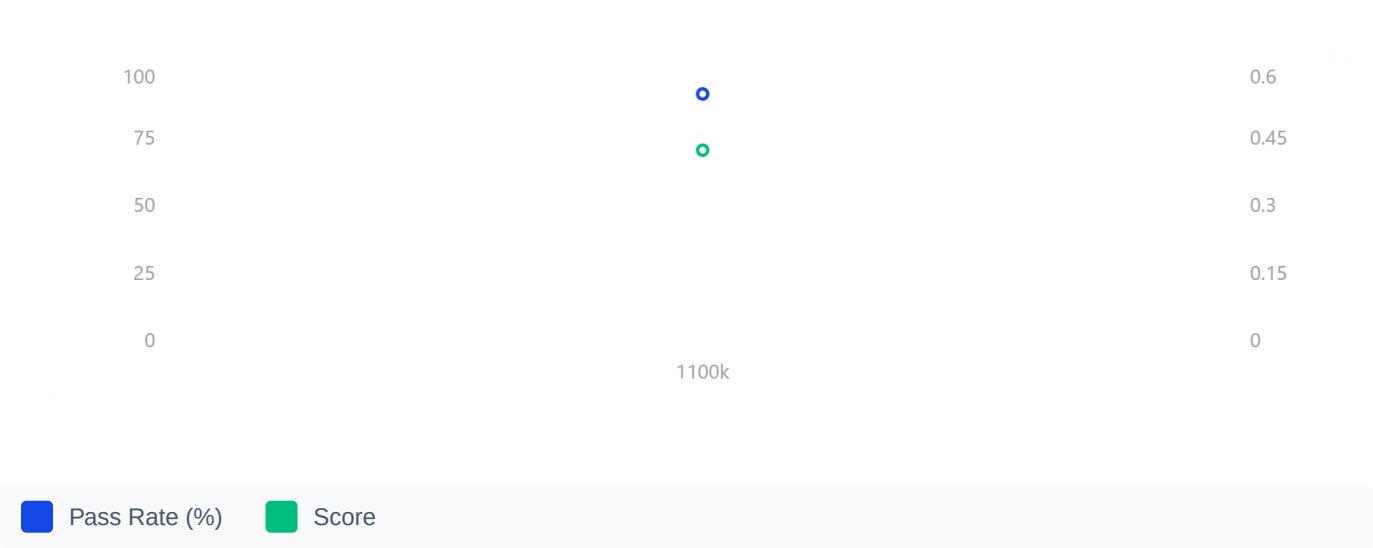
nist_linear_complexity



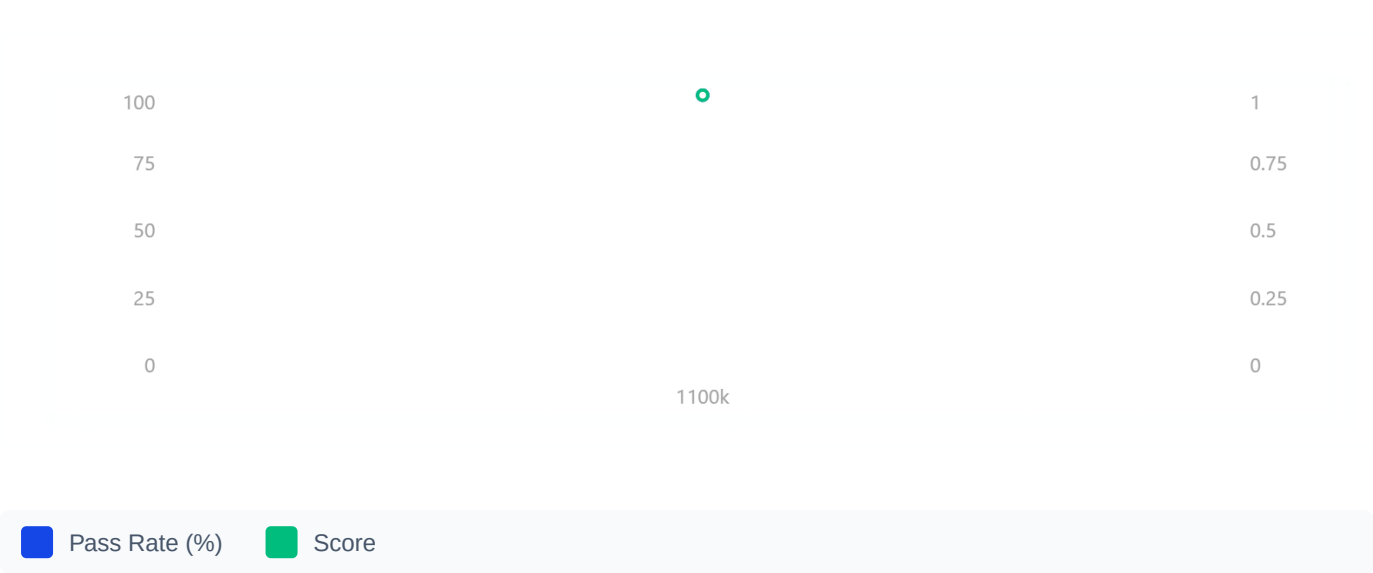
nist_universal



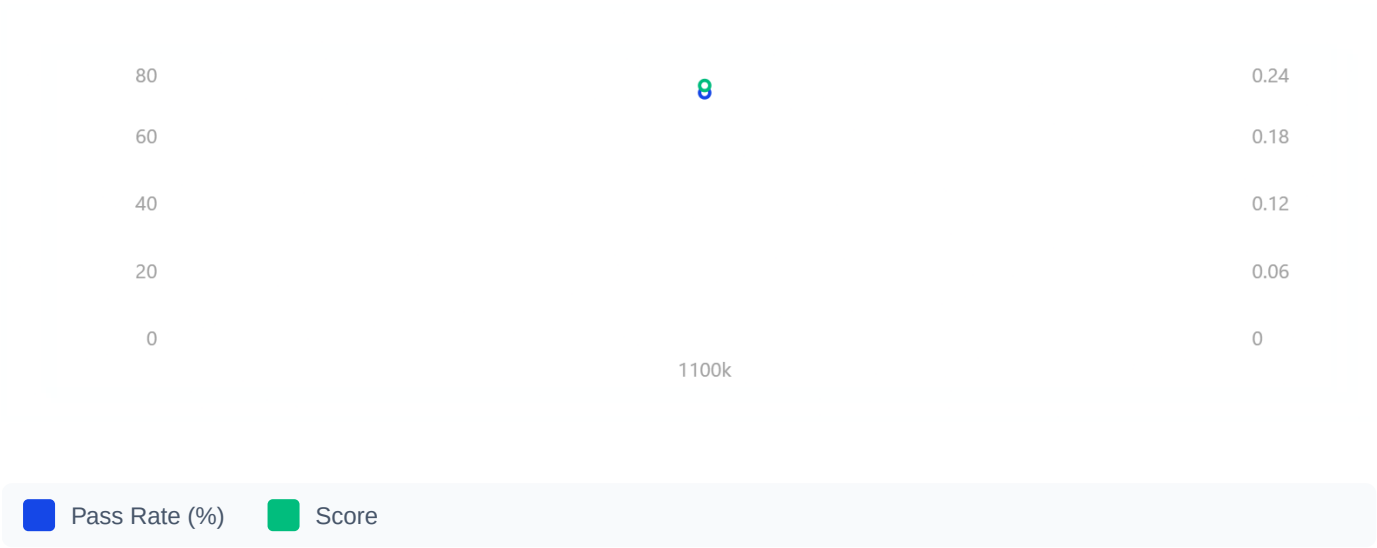
nist_overlapping_template



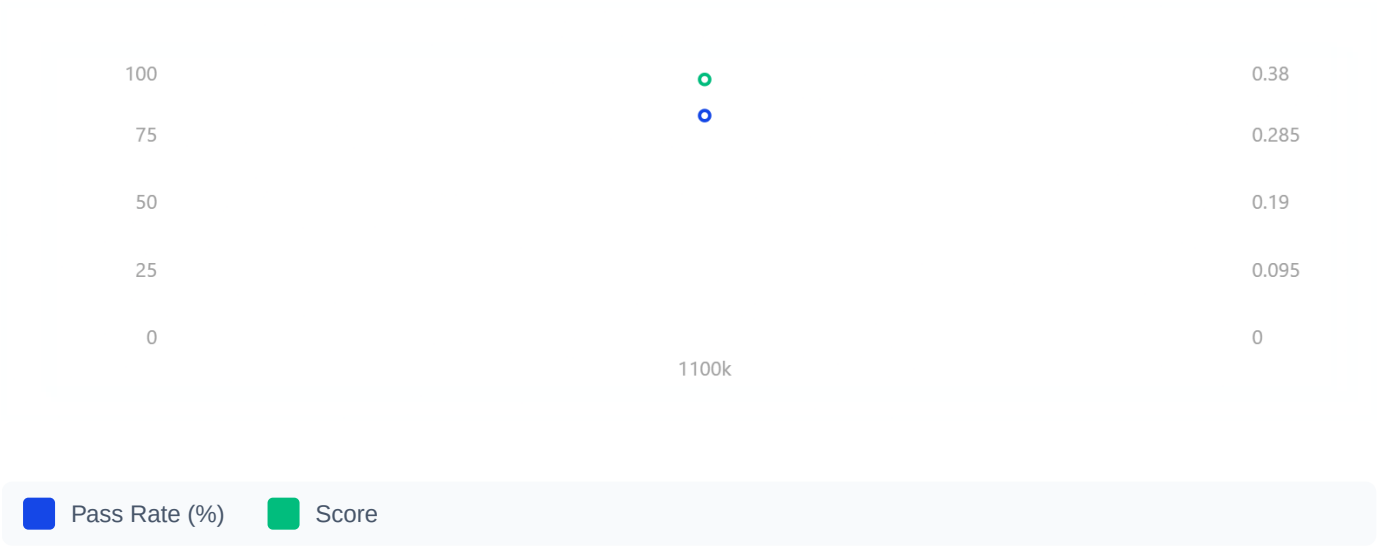
nist_non_overlapping_template



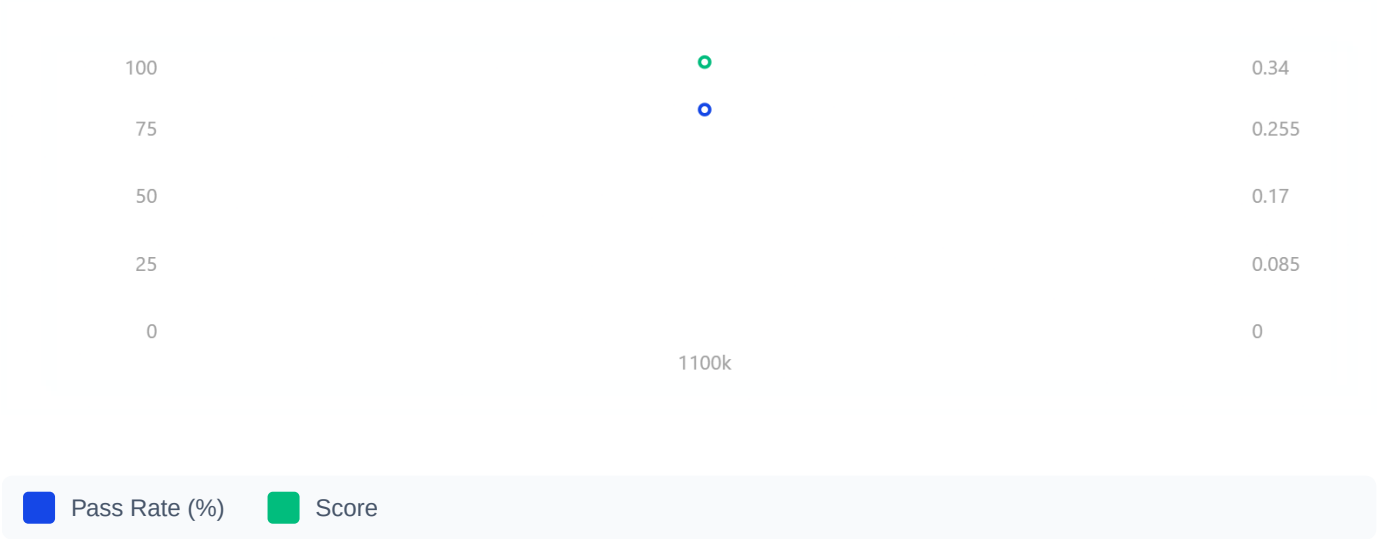
nist_dft



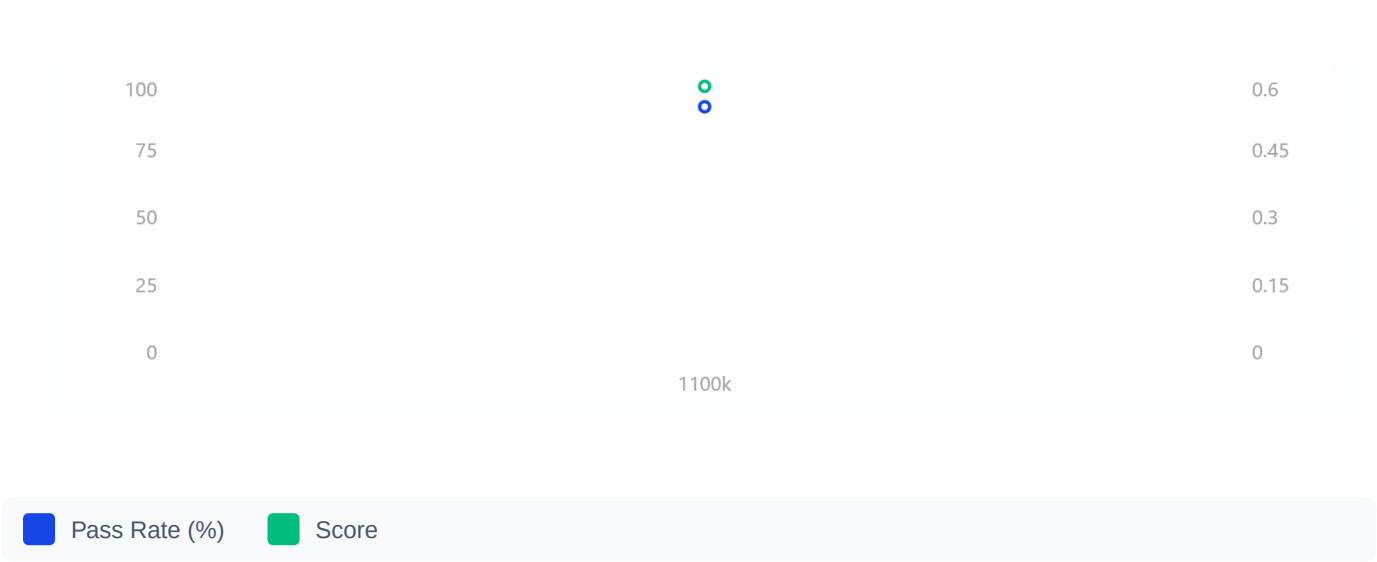
nist_matrix_rank



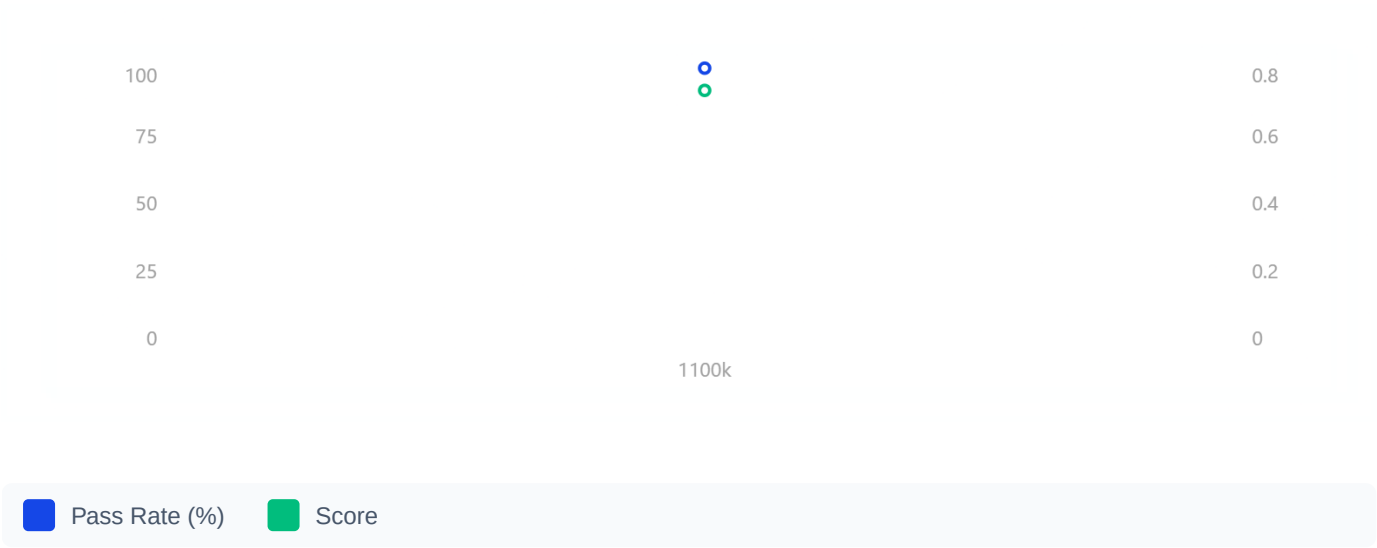
nist_longest_run



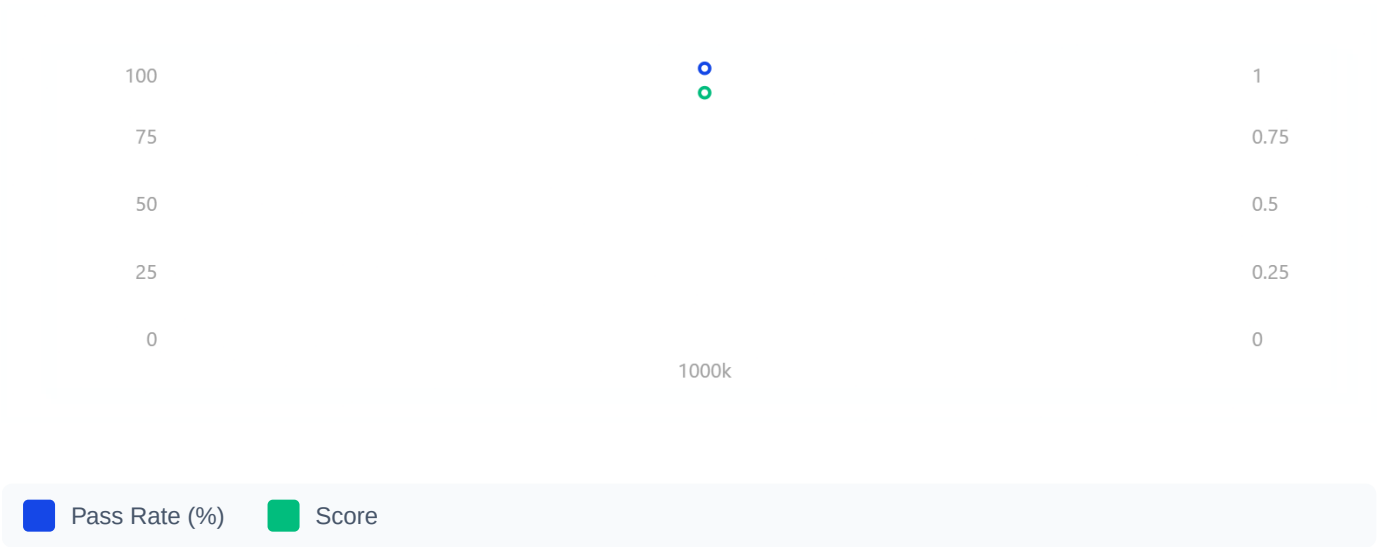
nist_runs



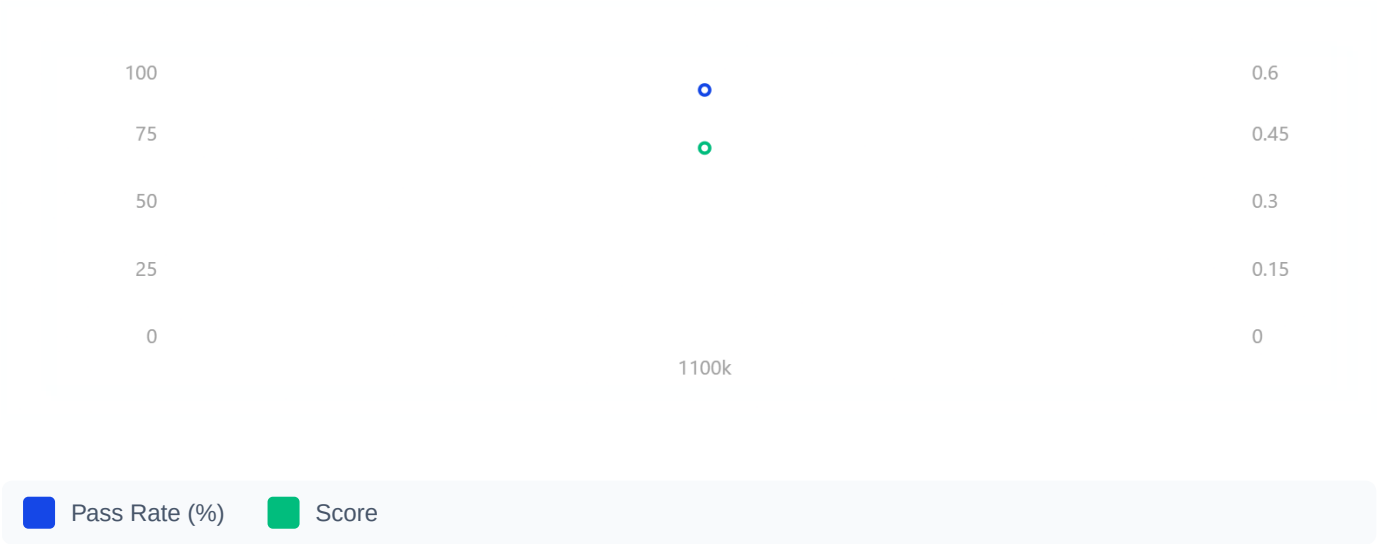
nist_block_frequency



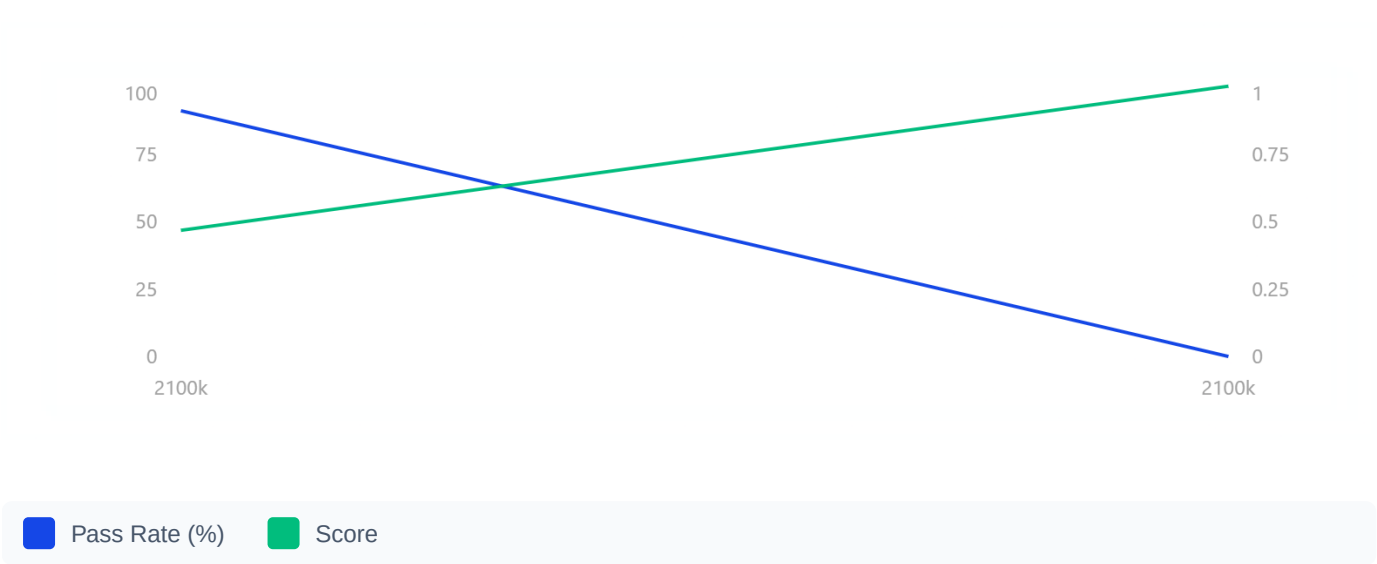
frequency_test



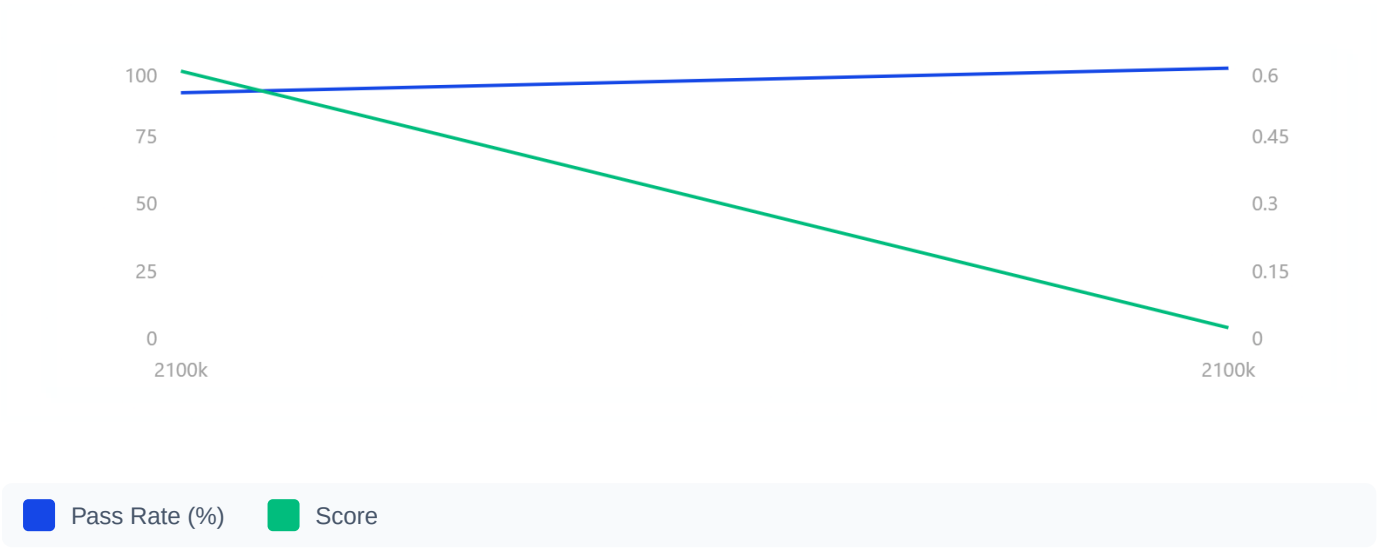
nist_monobit



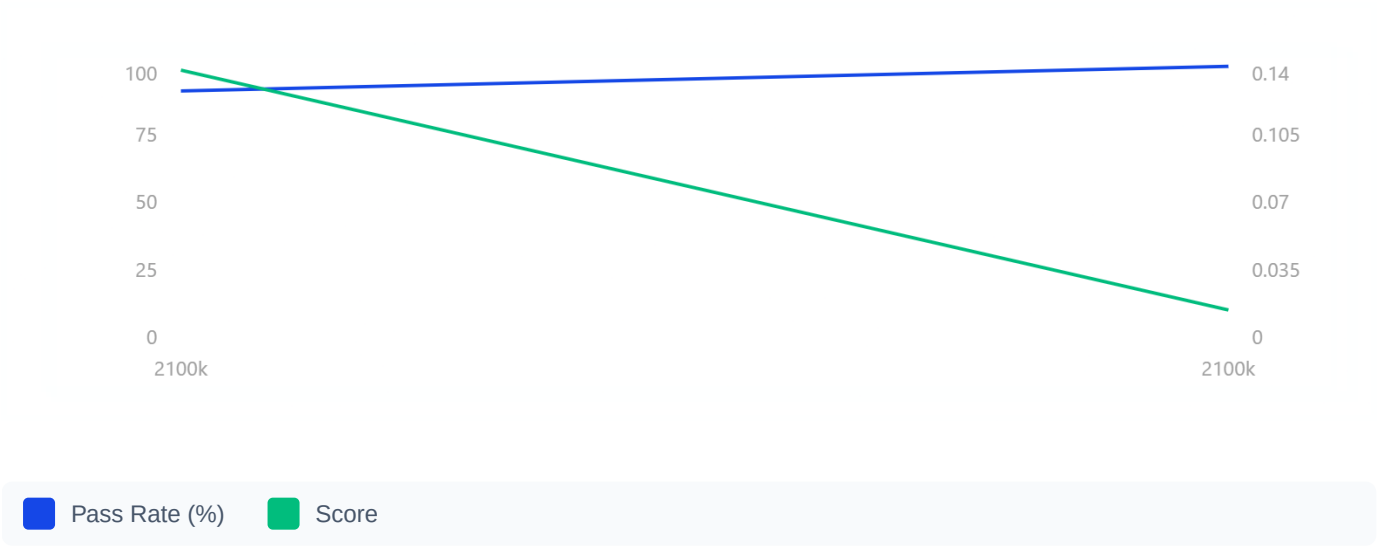
diehard_craps



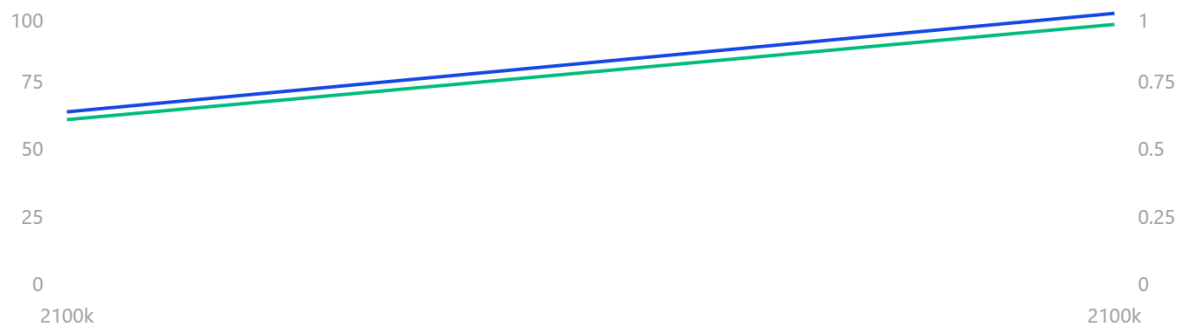
diehard_runs



diehard_overlapping_sums

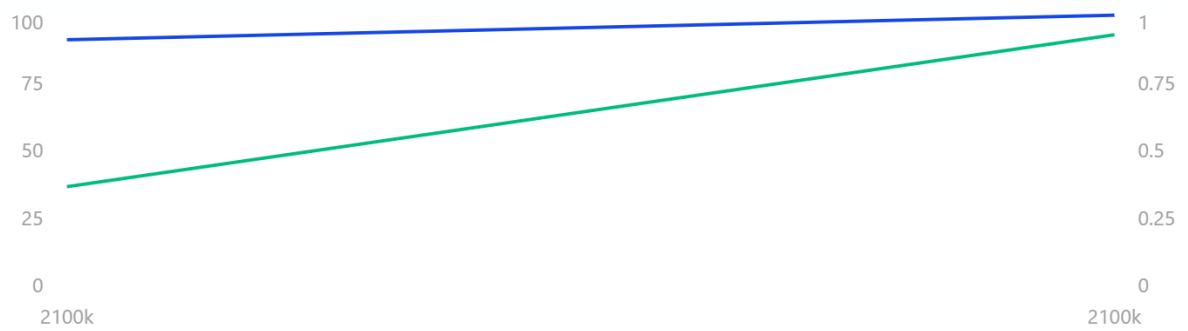


diehard_squeeze



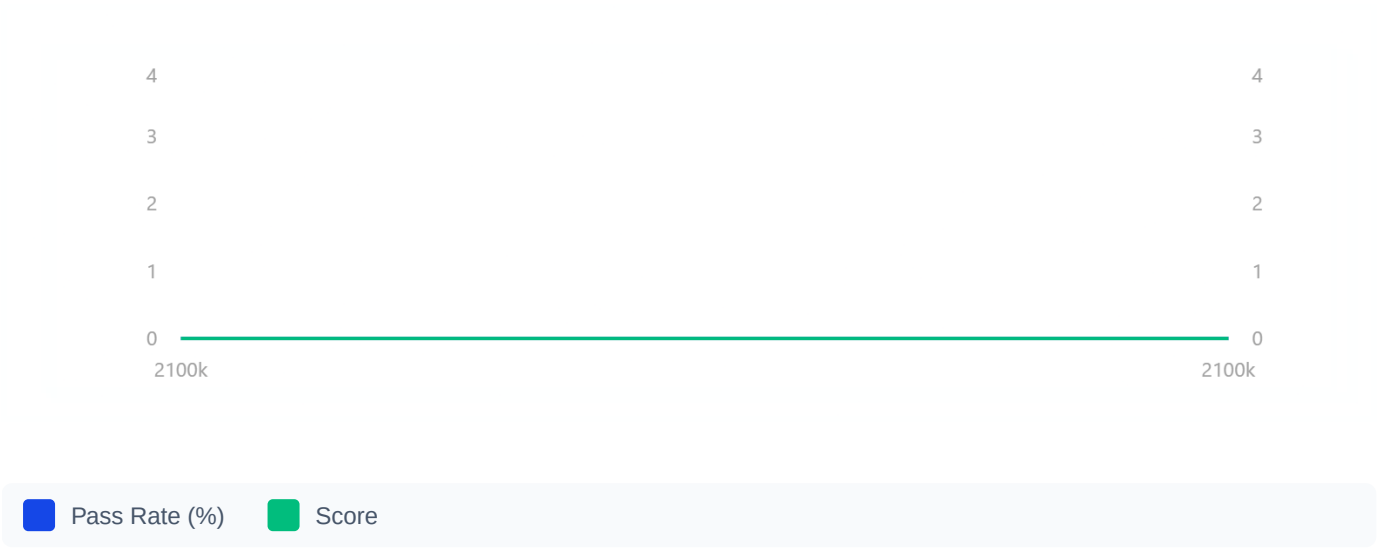
Pass Rate (%) Score

diehard_3dspheres

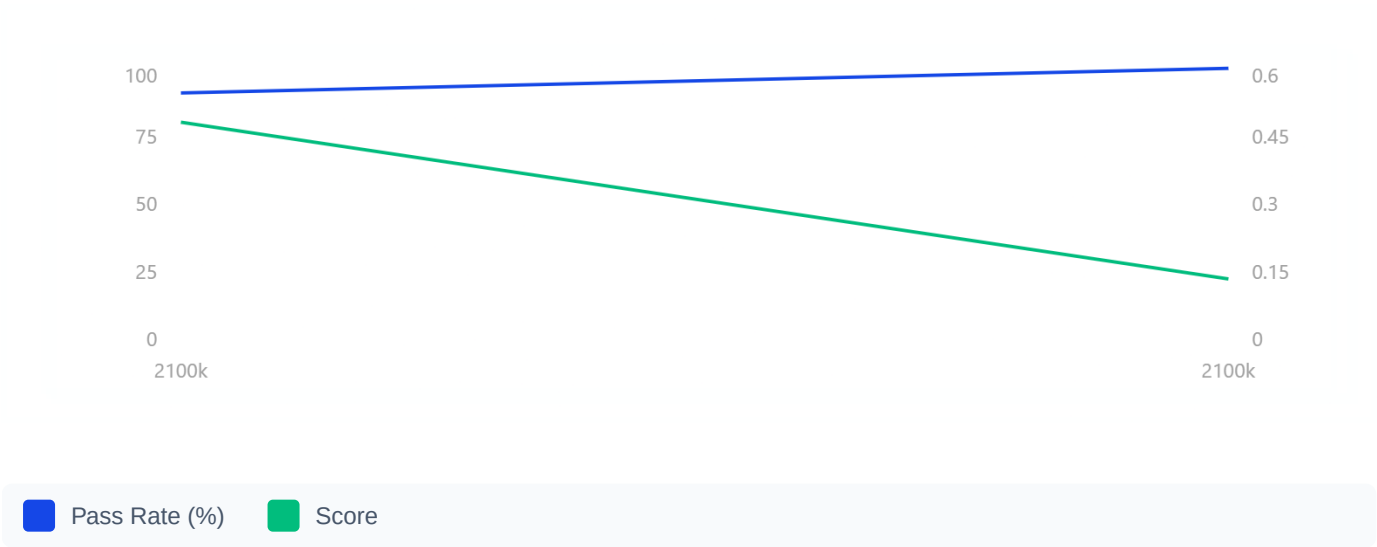


Pass Rate (%) Score

diehard_minimum_distance



diehard_parking_lot

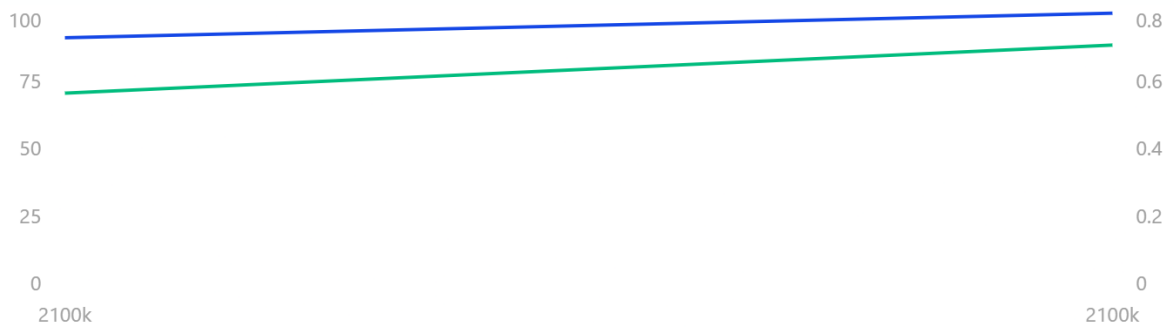


diehard_count_1s



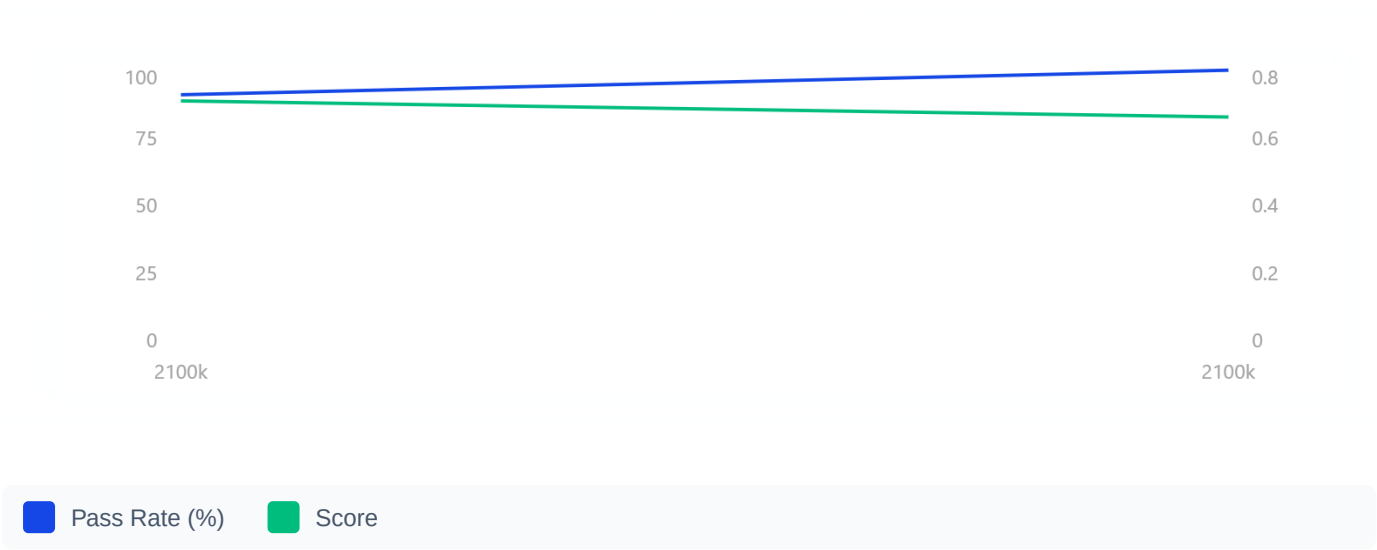
Pass Rate (%) Score

diehard_dna

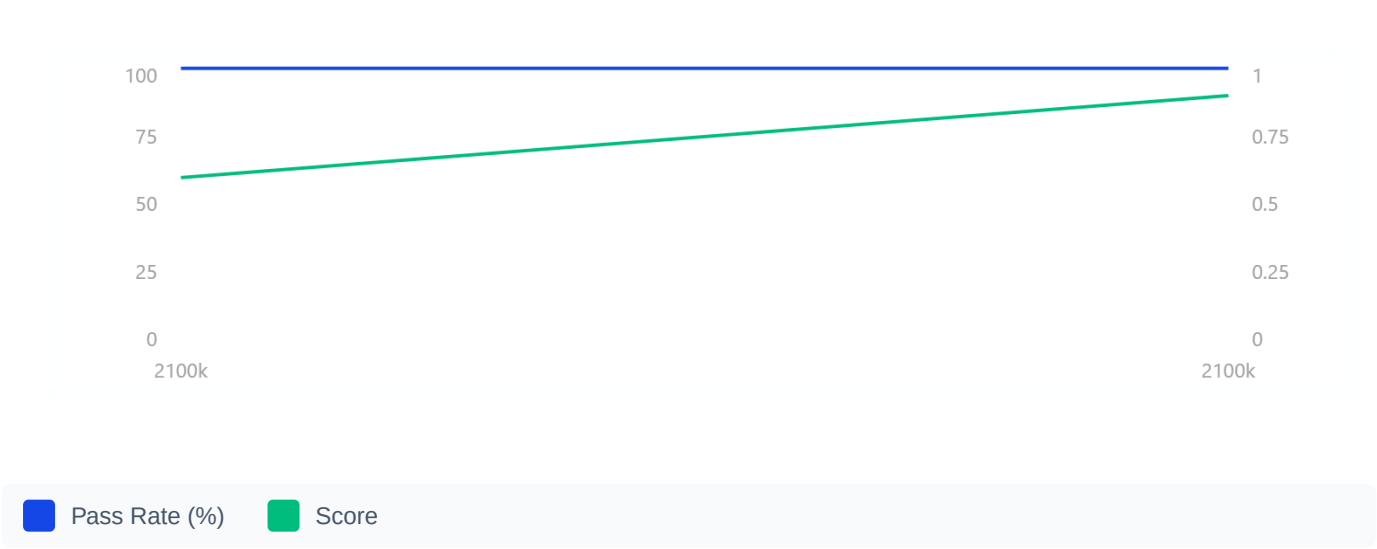


Pass Rate (%) Score

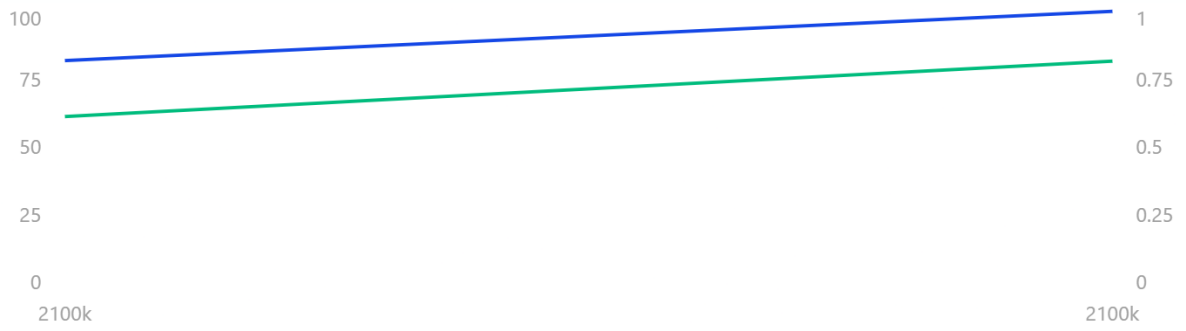
diehard_oqso



diehard_opso

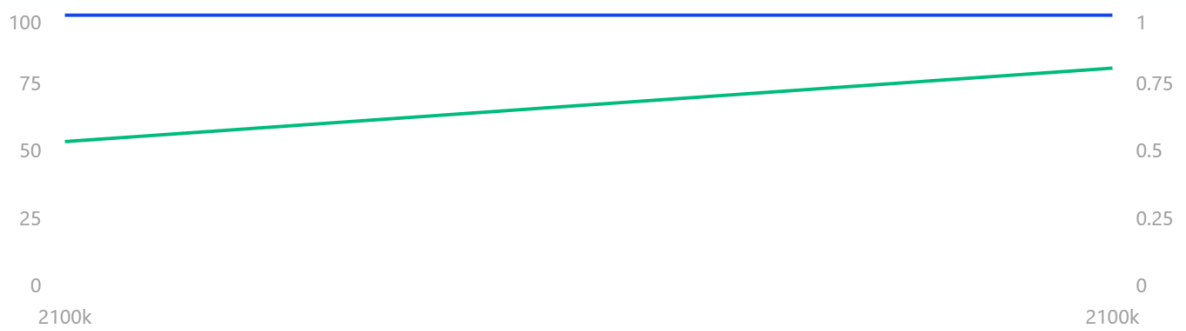


diehard_bitstream



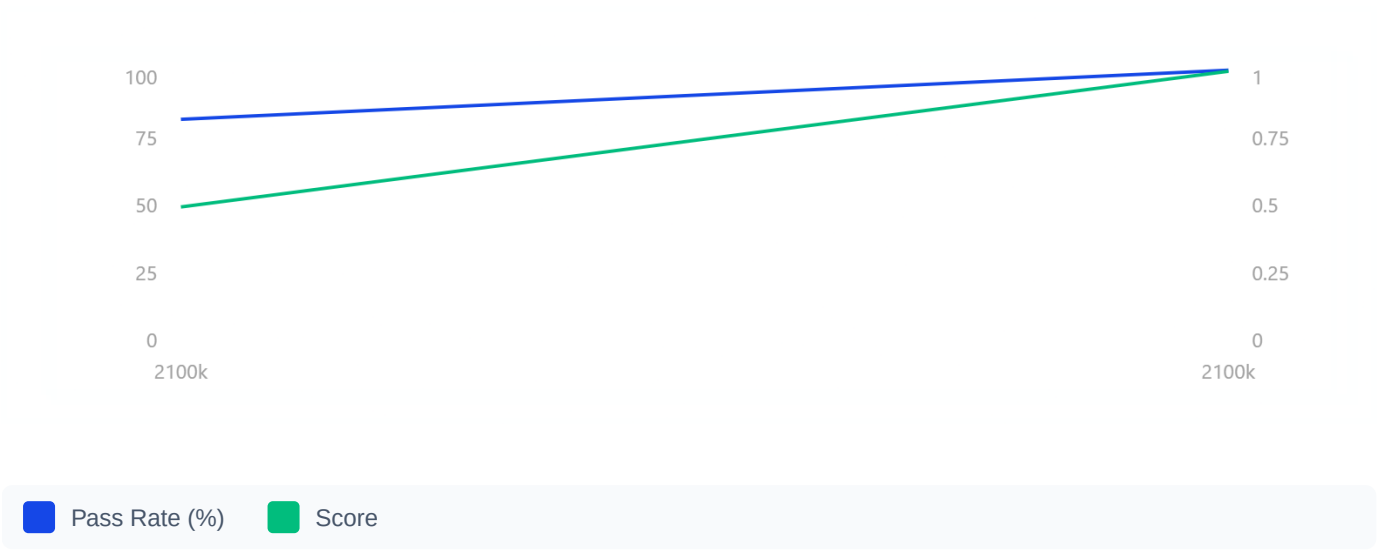
Pass Rate (%) Score

diehard_binary_rank



Pass Rate (%) Score

diehard_overlapping_permutations



diehard_birthday_spacings

